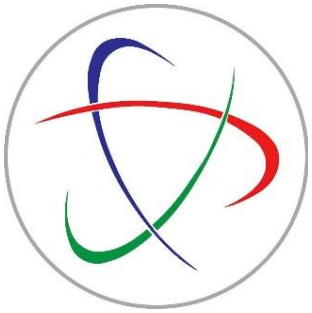
THE UNIVERSITY OF DANANG
UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF ELECTRONICS AND ARTIFICIAL INTELLIGENCE







PROJECT TITLE:

PROGRAMMING AND DEVELOPING A DIGITAL CLOCK DEVICE ON THE SN32F407_EVK BOARD

Team name: Chilli

Team members: Duong Quoc Bao
Pham Huynh Duc
Huynh Phuc Thien
Nguyen Duc Anh Tuan

Da Nang, August/2026

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1 Design proposal

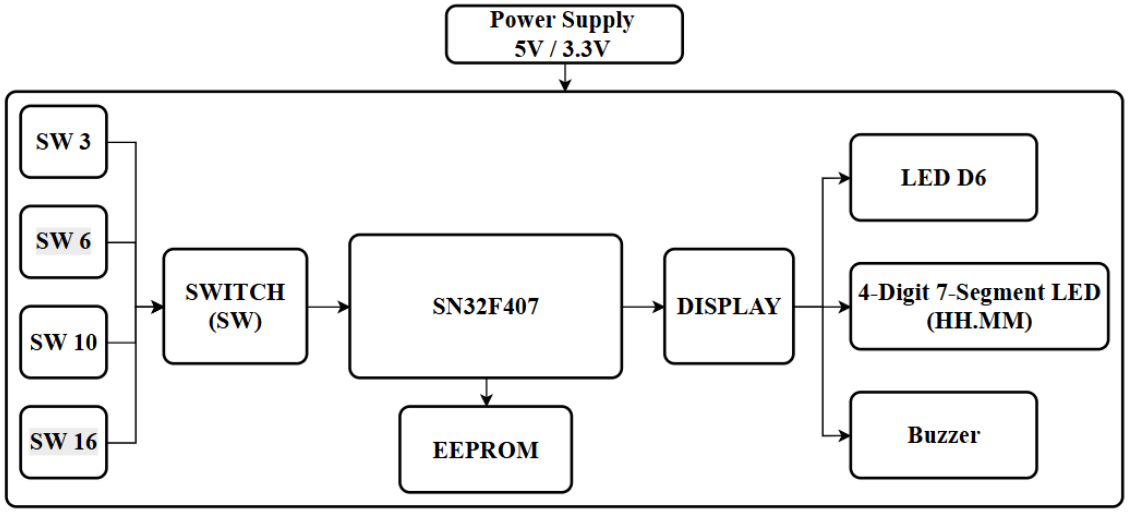
The digital clock and timer system is built around the SN32F407 microcontroller, which serves as the central processing unit for time management and peripheral control. From a software perspective, the system operates using a "Superloop" architecture combined with a continuous polling technique. Following initial hardware initialization, the microcontroller continuously executes loops to update real-time data, scan button states, and output information to the display.

Time is visually presented on an LED display in HH:MM format. To manage these functions smoothly, the system employs a Finite State Machine (FSM) approach. Users can toggle between modes standard time display, real-time adjustment, and timer setting using push-buttons. In setting mode, the active data field flashes for easy identification, while the FSM ensures the system accurately distinguishes which parameter is being adjusted (incremented or decremented) without operational conflicts.

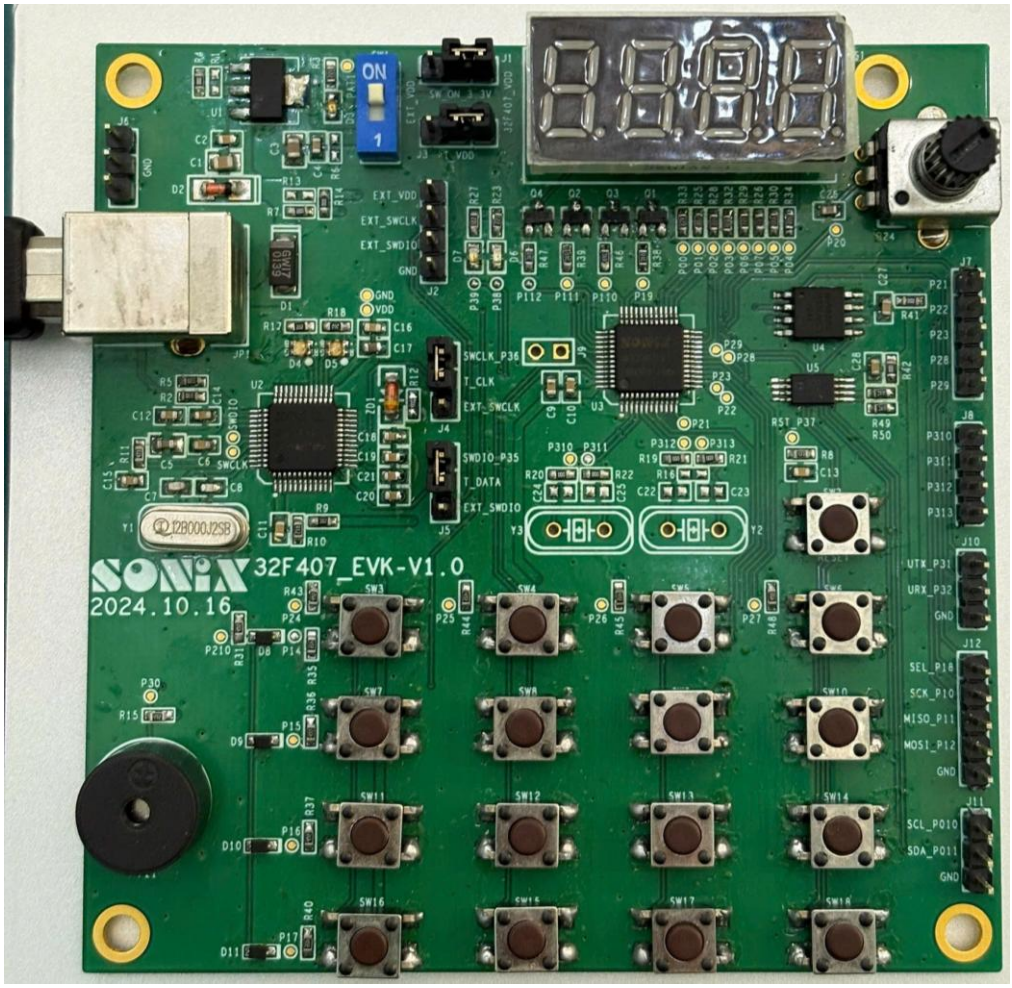
To ensure simultaneous functionality, the program utilizes a non-blocking timing technique with a 1-millisecond base interval. Instead of using blocking commands that would freeze system operations, tasks such as timekeeping, indicator blinking, and button response are assigned distinct cycles and executed in parallel. This results in highly accurate timekeeping and near-instantaneous user input response.

Finally, once configured, the timer setting is tagged with a "Magic Byte" validation marker and securely stored in EEPROM memory, preventing data loss during power outages. During operation, the system continuously compares the current time with the set time; when the values match, a buzzer is immediately activated to sound an alert. Furthermore, to optimize stability, the system is equipped with an automatic timeout mechanism that exits setup mode after 30 seconds of inactivity, combined with a Watchdog Timer feature that automatically restarts the circuit in the event of a system freeze.

2 System block diagram



3 Board SONIX 32F407_EVK-V1.0



3.1 Central Processing Unit - MCU (SONIX SN32F407FG)



Board location: It is a large, square, black IC chip located in the central area (slightly to the right) of the circuit board, featuring the faintly printed text "SONIX".

Functions and Duties:

- Acts as the "brain" controlling the entire system.
- Generates a real-time counter based on the system scan cycle (ms_count). Every 1000ms (1 second), it updates the `second` variable and automatically calculates `minute` and `hour` values (hours cycle from 0–23; minutes from 0–59).
- Continuously scans push-button states, sends data to the LED display, performs read/write operations with the EEPROM, and controls the buzzer.
- Integrates a Watchdog Timer (WDT) for system protection; the MCU automatically restarts if the program hangs.

3.2 Display (7-segment LED display)



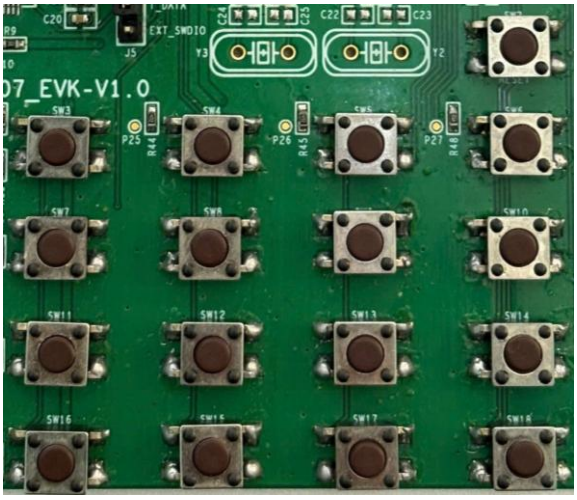
Board location: The prominent red display module is in the top-right corner. In the image, the display shows the value "00.00".

Functions and Duties:

- Displays time information to the user in HH.MM (Hour.Minute) format.

- Provides visual feedback via a flashing effect:
- When the user enters MODE_SET_HOUR or MODE_ALARM_HOUR: the "Hour" digits flash (0.5s on / 0.5s off cycle).
- When entering MODE_SET_MIN or MODE_ALARM_MIN: the "Minute" digits flash.

3.3 Push-Button Matrix



Board location: Switches SW3, SW6, SW10, and SW16 are part of the switch matrix located on the underside of the board.

Functions and Duties:

- Button SW3 (Settings / MODE): Switches the clock's operating mode. Press to cycle through: Normal Operation → Set Hour → Set Minute → Return to Normal.
- Button SW16 (Timer / ALARM): Sets the alarm. Press to cycle through: Normal → Set Alarm Hour → Set Alarm Minute → Save to EEPROM & enable alarm.
- Button SW6 (INCREASE / +): Increases the hour or minute value by 1 unit while in setting mode. Upon reaching the limit (24 hours or 60 minutes), it automatically wraps around to 0.
- Button SW10 (DECREASE / -): Decreases the hour or minute value by 1 unit. If at 0, decreasing further wraps around to the maximum value (23 hours or 59 minutes).

3.4 Buzzer



Board location: A black cylindrical object with a hole in the center, located in the bottom-left corner.

Functions and Duties:

- Alarm (Buzzer): When the clock reaches the preset time (hours, minutes, and seconds), the buzzer emits a continuous "Beep-Beep" sound for 5 seconds (alternating 0.5s on / 0.5s off).
- Operation Feedback (Buzzer): Whenever buttons SW3, SW6, SW10, or SW16 are pressed, or upon automatically exiting setup mode (after a 30-second timeout), the buzzer emits a short "Beep" (0.3s) to provide confirmation.

3.5 Status Indicator LED (D6)



Board location: Located below the four 7-segment LEDs and offset to the left.

Functions and Duties:

- Flashes continuously with a 1-second cycle (0.5s on – 0.5s off) when the user is in any Setting or Timer mode. This light remains off in normal timekeeping mode.

3.6 Data Storage Module (EEPROM)

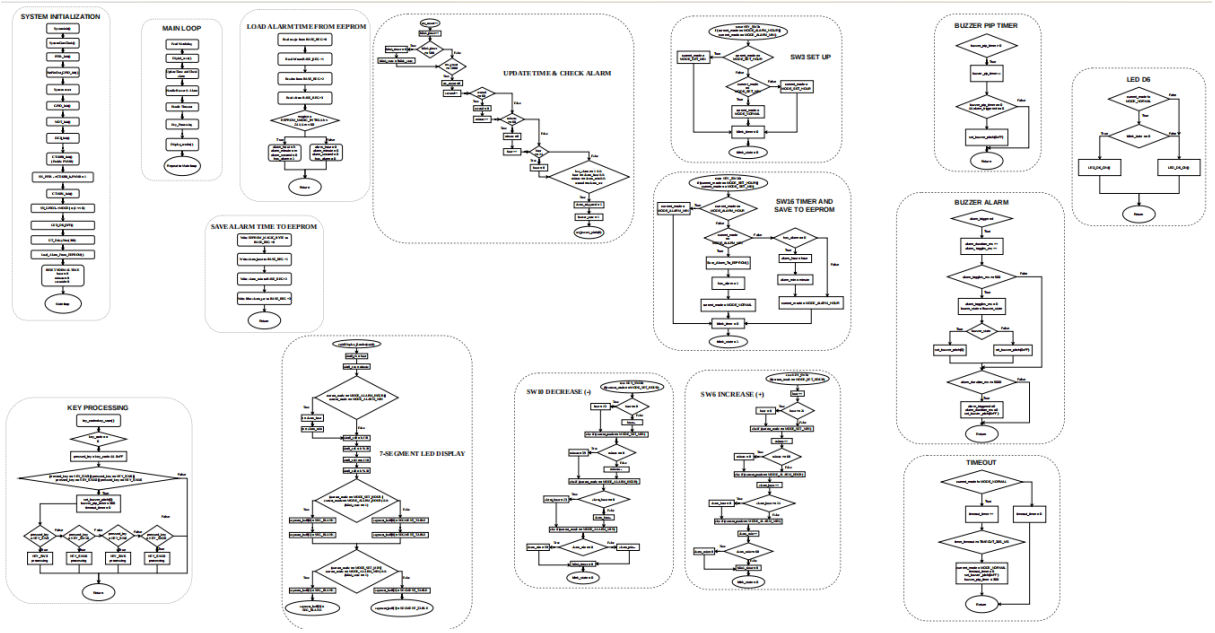


Board location: It is a small 8-pin IC (U4) located beneath the four 7-segment LEDs and U5, towards the right side.

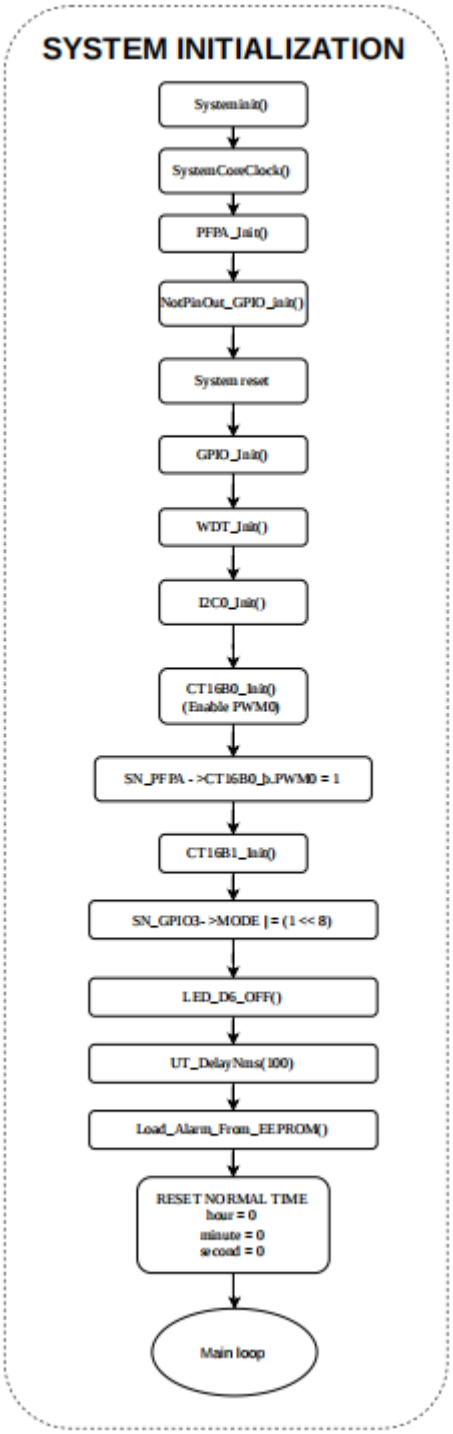
Functions and Duties:

- Permanent storage: Responsible for retaining the alarm time (hours and minutes) set by the user.
- Data recovery: Upon power loss or system reset, the `Load_Alarm_From_EEPROM()` function scans the chip. If the correct "Magic Byte" (0xAA) is detected, it automatically restores the previously set alarm time, eliminating the need for the user to reconfigure it.

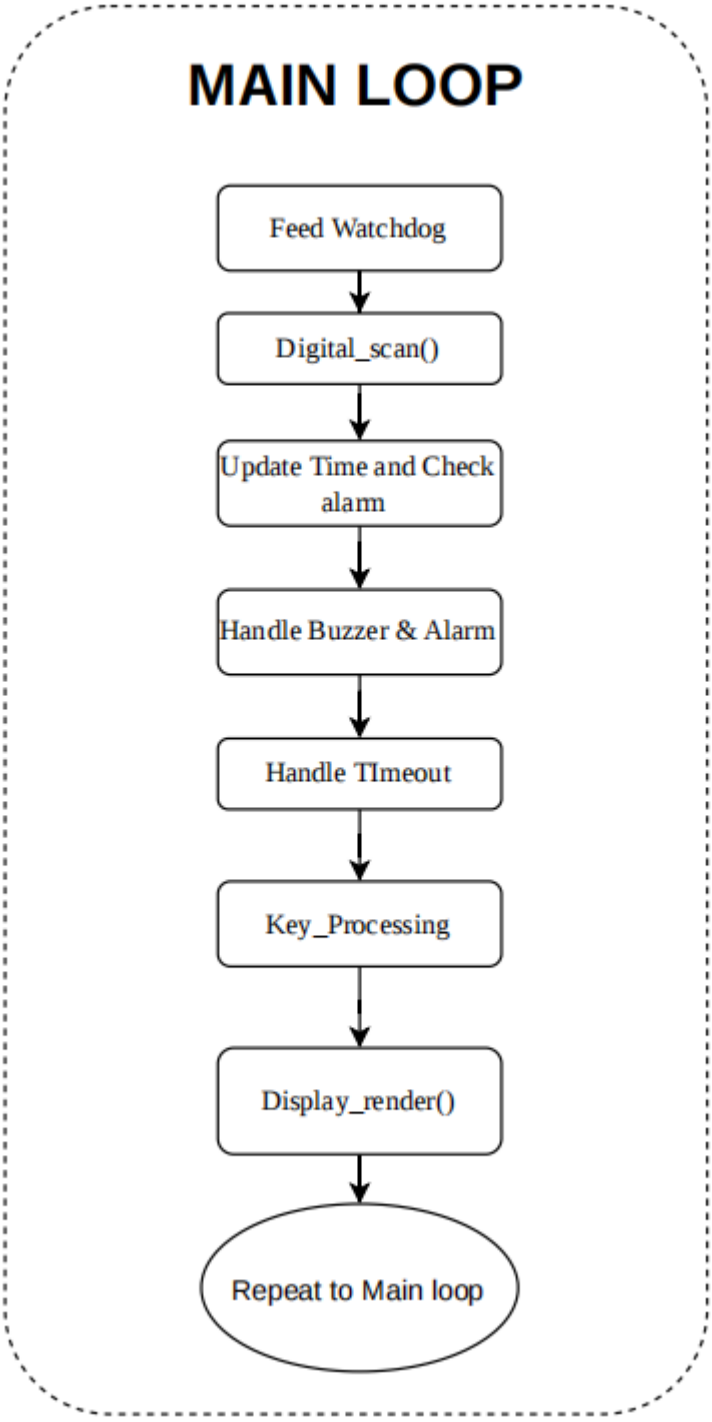
4 Algorithm Flowchart



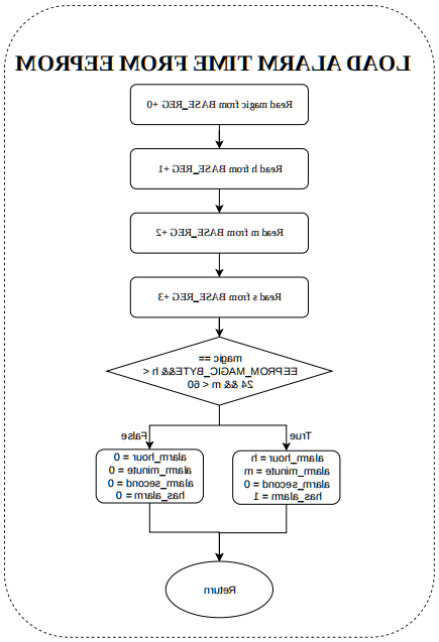
4.1 System Initialization

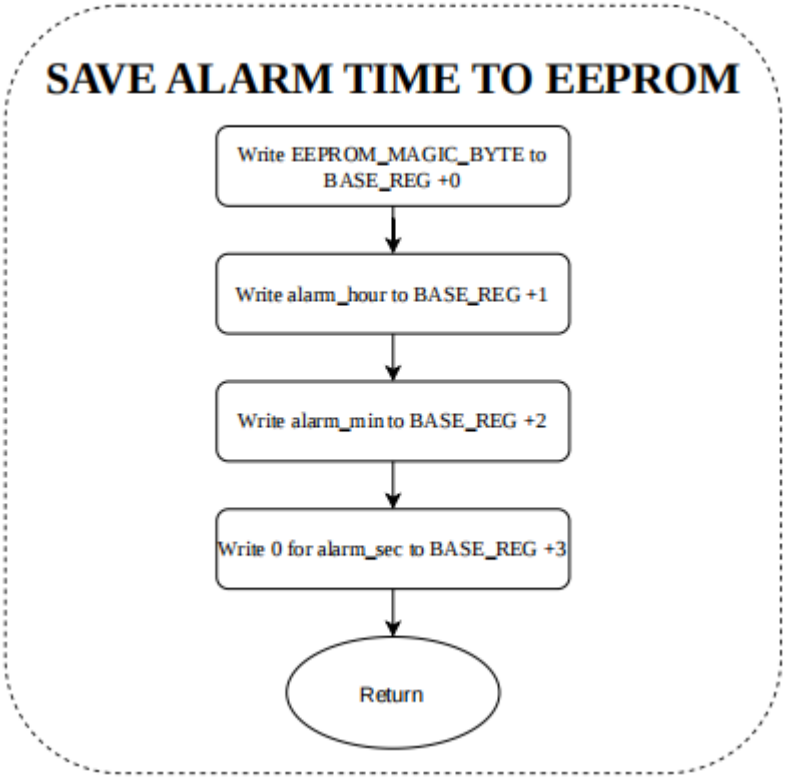


4.2 Main Loop

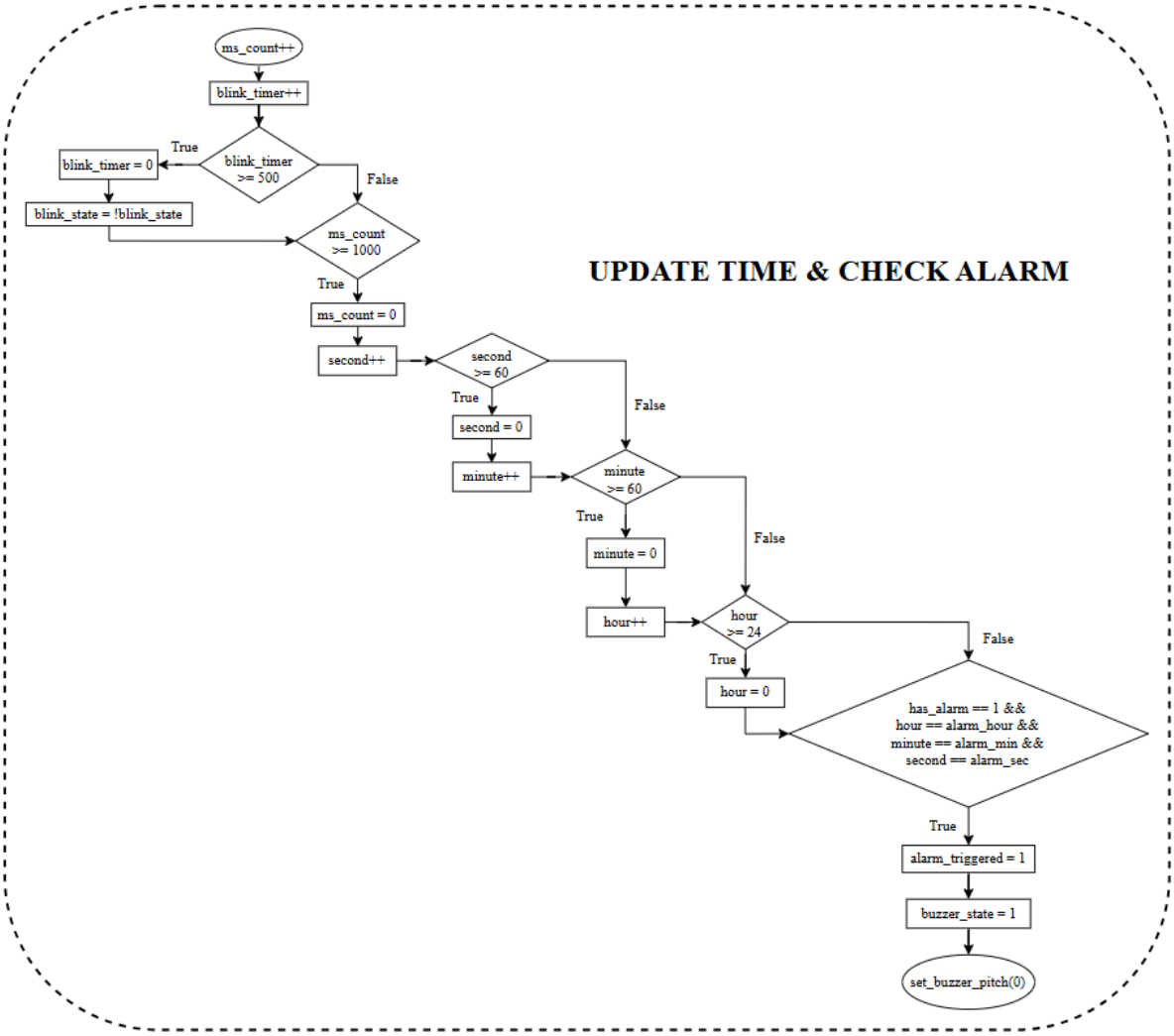

MAIN LOOP
Feed Watchdog
Digital_scan()
Update Time and Check alarm
Handle Buzzer & Alarm
Handle TImeout
Key_Processing
Display_render()
Repeat to Main loop

4.3 EEPROM

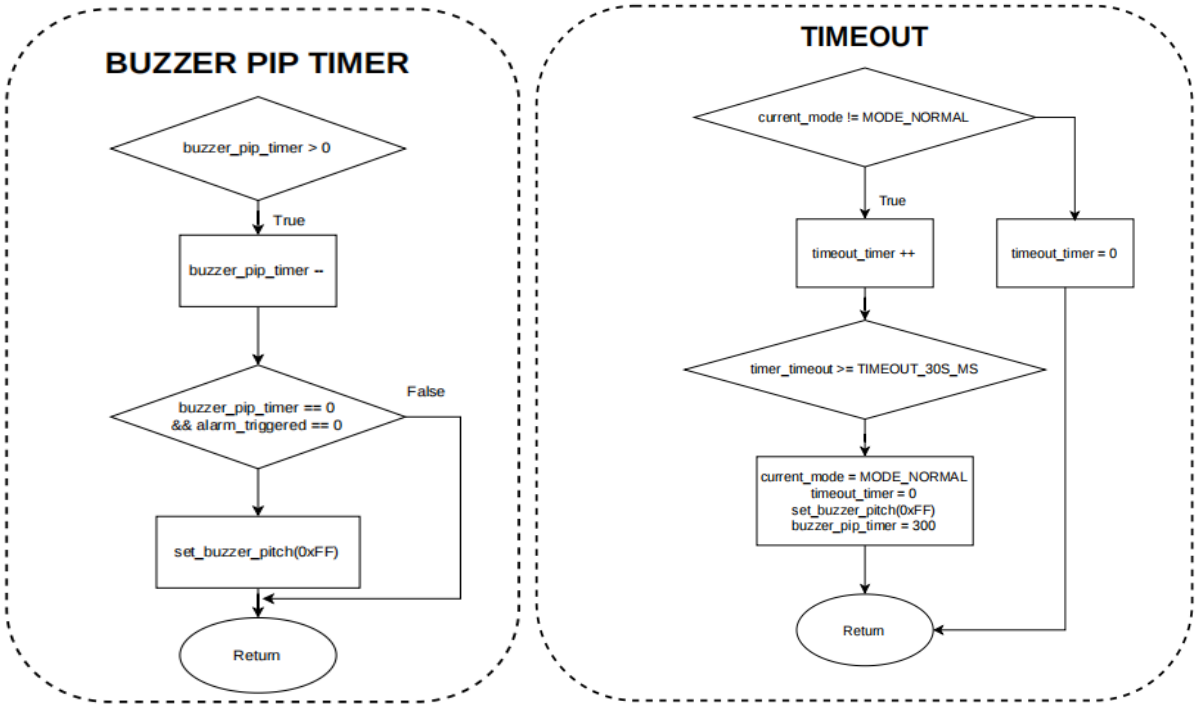


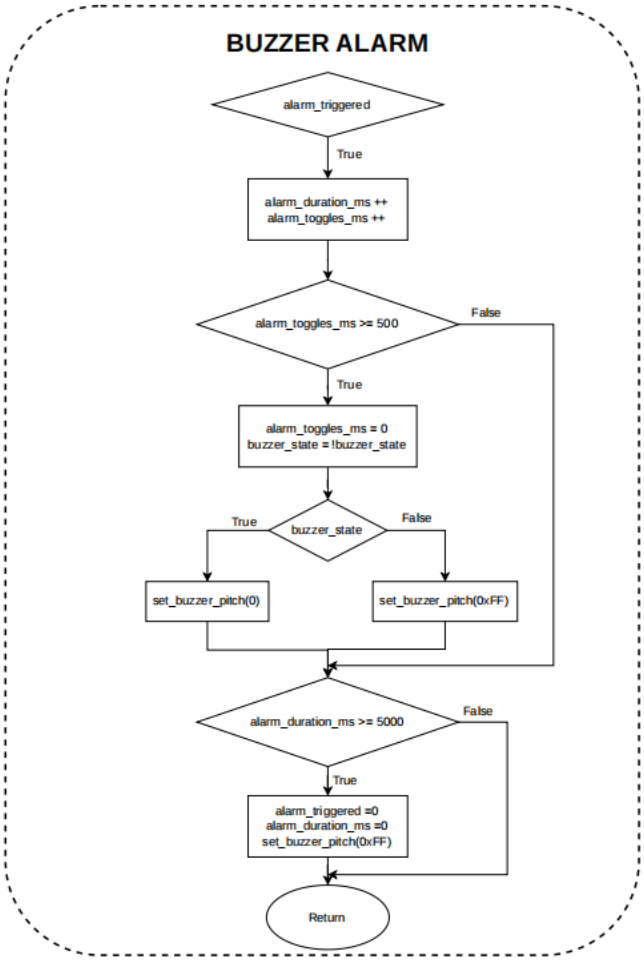

SAVE ALARM TIME TO EEPROM
Write EEPROM_MAGIC_BYTE to BASE_REG +0
Write alarm_hour to BASE_REG +1
Write alarm_min to BASE_REG +2
Write 0 for alarm_sec to BASE_REG +3
Return

4.4 Update Time & Check Alarm

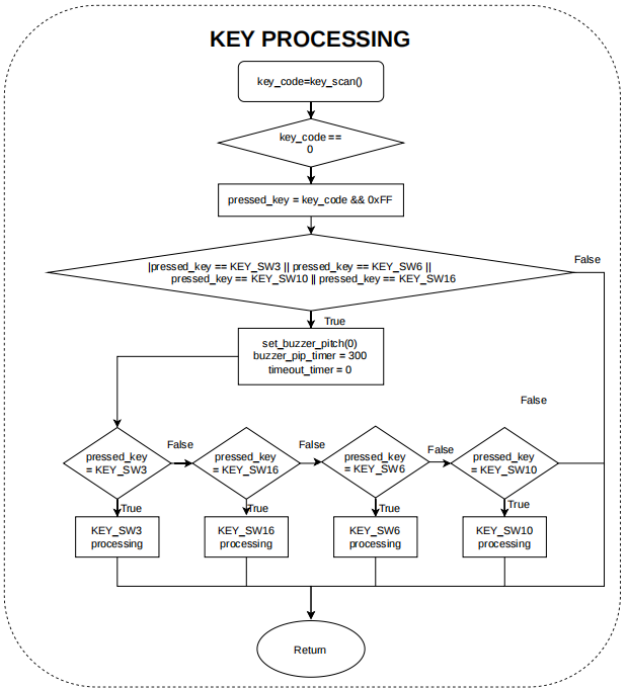


4.5 Buzzer and timeout

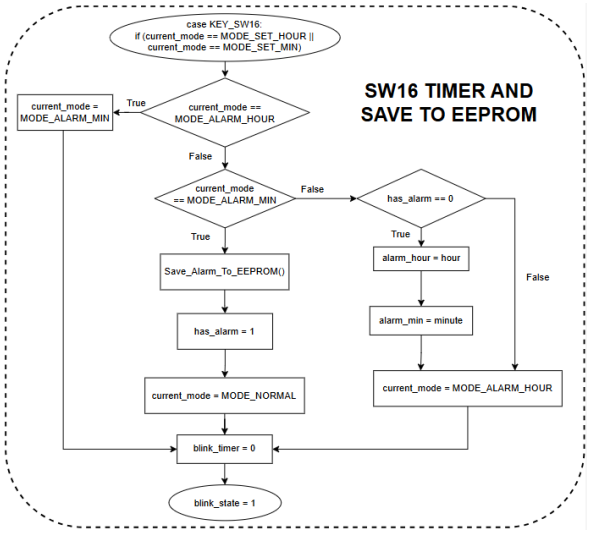


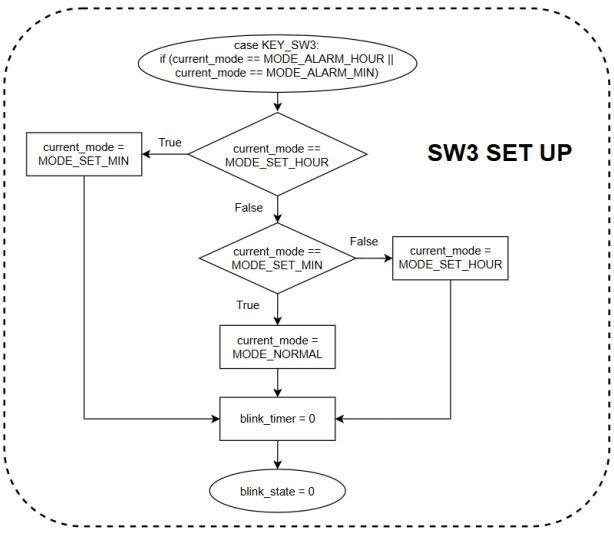


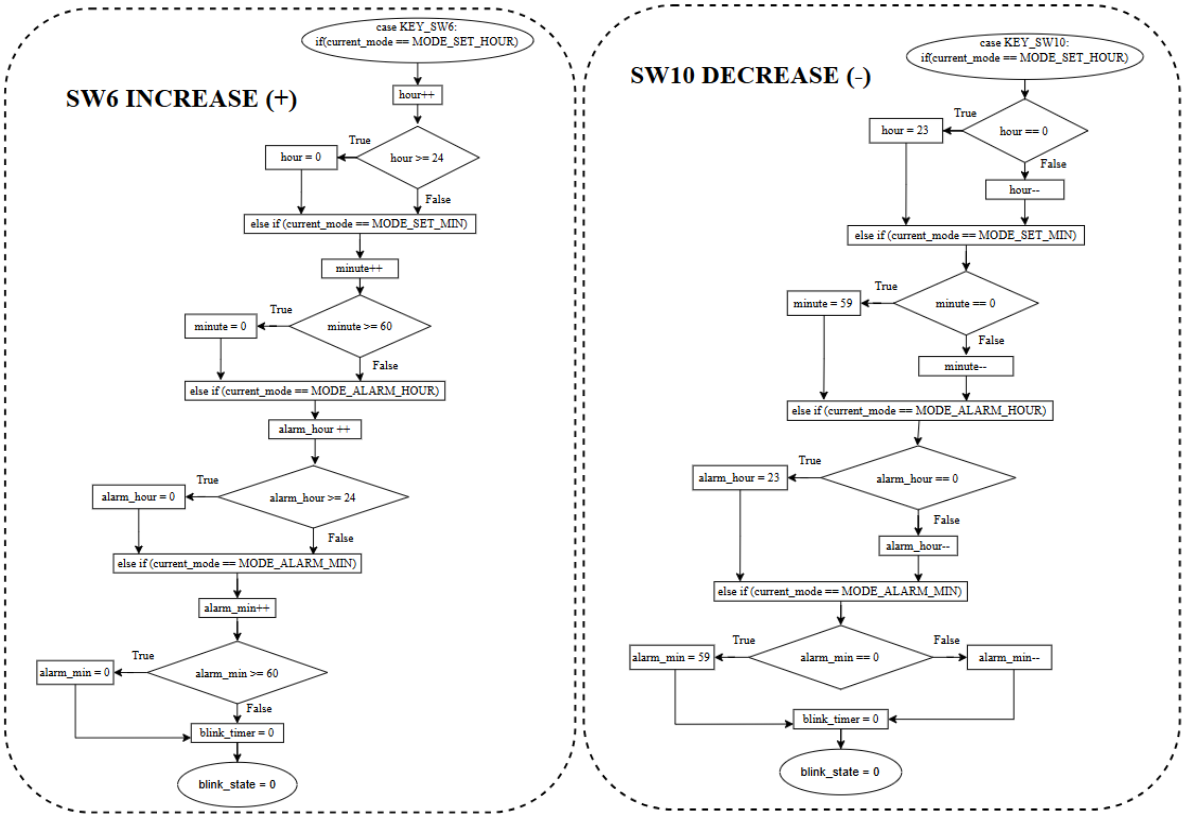
4.6 Button Processing



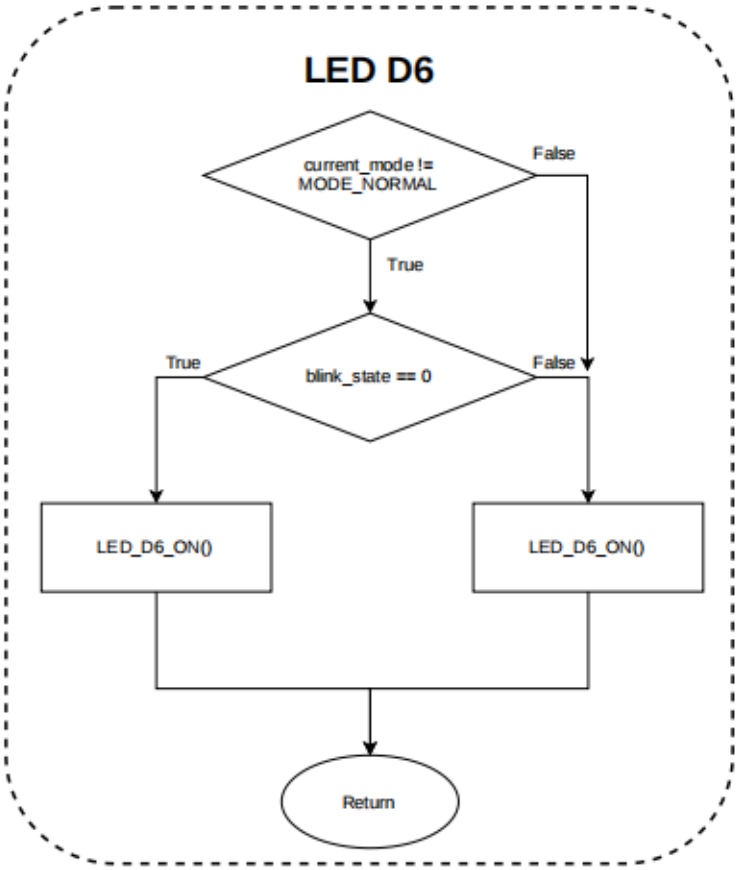
For each key:



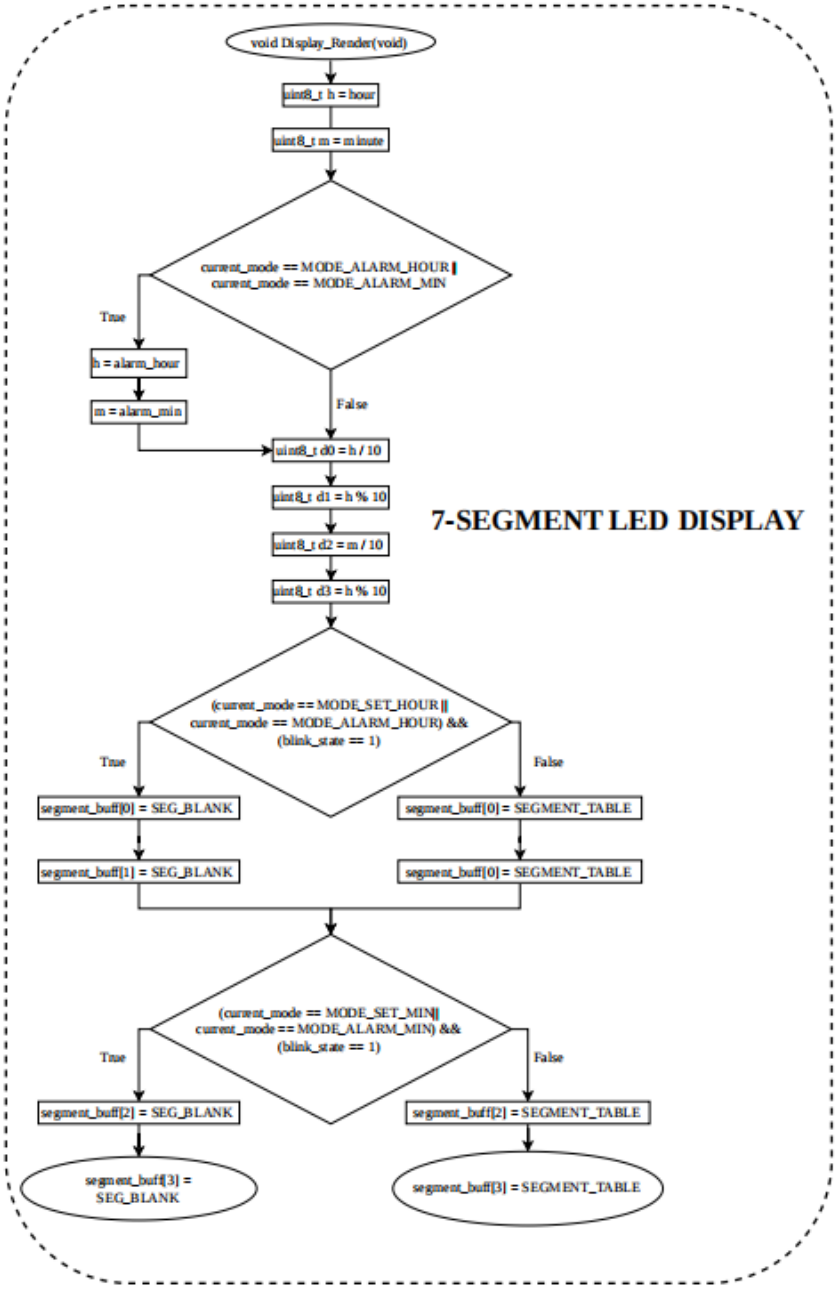



SW6 INCREASE (+)
case KEY_SW6:
if(current_mode == MODE_SET_HOUR)
hour++
hour >= 24
True
hour = 0
False
else if (current_mode == MODE_SET_MIN)
minute++
minute >= 60
True
minute = 0
False
else if (current_mode == MODE_ALARM_HOUR)
alarm_hour ++
alarm_hour >= 24
True
alarm_hour = 0
False
else if (current_mode == MODE_ALARM_MIN)
alarm_min++
alarm_min >= 60
True
alarm_min = 0
False
blink_timer = 0
blink_state = 0
SW10 DECREASE (-)
case KEY_SW10:
if(current_mode == MODE_SET_HOUR)
hour == 0
True
hour = 23
False
hour--
else if (current_mode == MODE_SET_MIN)
minute == 0
True
minute = 59
False
minute--
else if (current_mode == MODE_ALARM_HOUR)
alarm_hour == 0
True
alarm_hour = 23
False
alarm_hour--
else if (current_mode == MODE_ALARM_MIN)
alarm_min == 0
True
alarm_min = 59
False
alarm_min--
blink_timer = 0
blink_state = 0

4.7 Led D6


LED D6
current_mode != MODE_NORMAL
False
True
blink_state == 0
True
False
LED_D6_ON()
LED_D6_ON()
Return

4.8 7-Segment Display

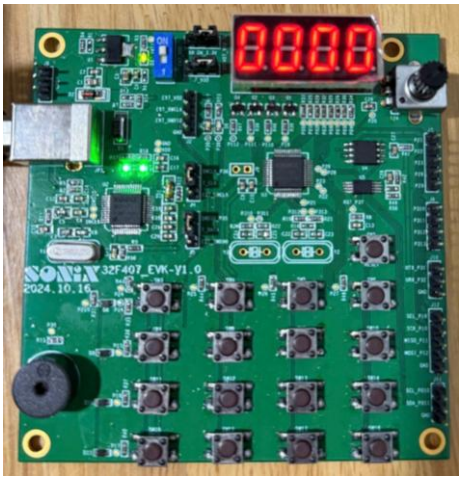


5 Implementation results

Requirement 1: FOUR 7-SEGMENT LEDS DISPLAY HOURS AND MINUTES IN NORMAL MODE USING THE FORMAT HH.MM.

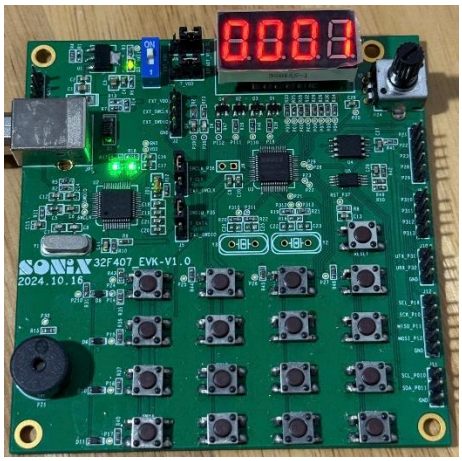
- Default display: Use four 7-segment LEDs to display hours and minutes in the HH.MM format.

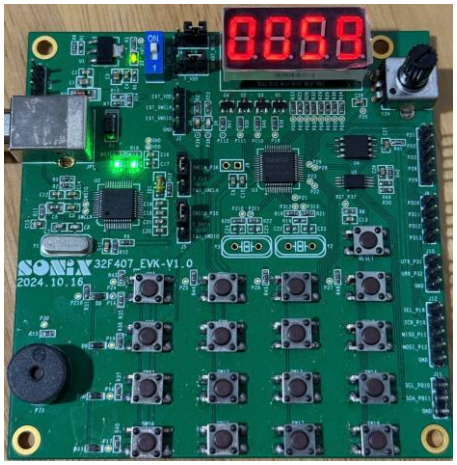
- Startup: Upon power-up, the circuit operates in normal mode, with hours and minutes reset to 00:00.

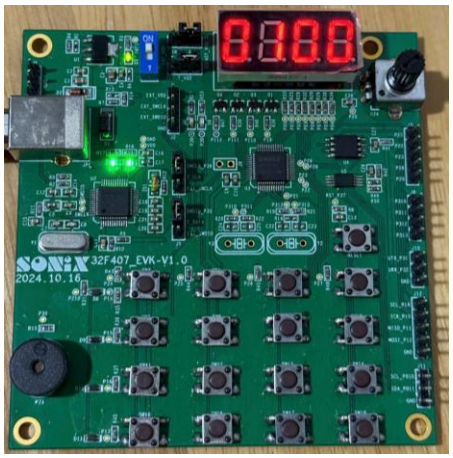


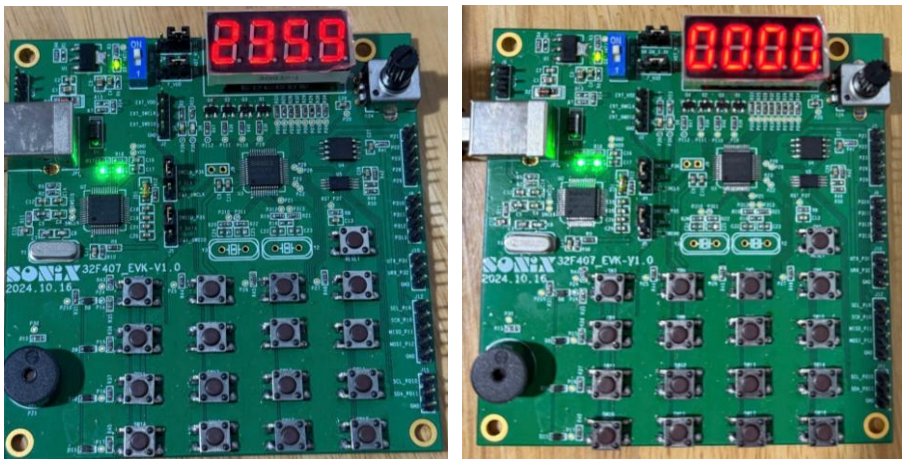
- Timer: The minute count increases by 1 every minute. Upon reaching the 60th minute, the minutes reset to 0 and the hour increases by 1. The hour is displayed in the range of 00 to 23.





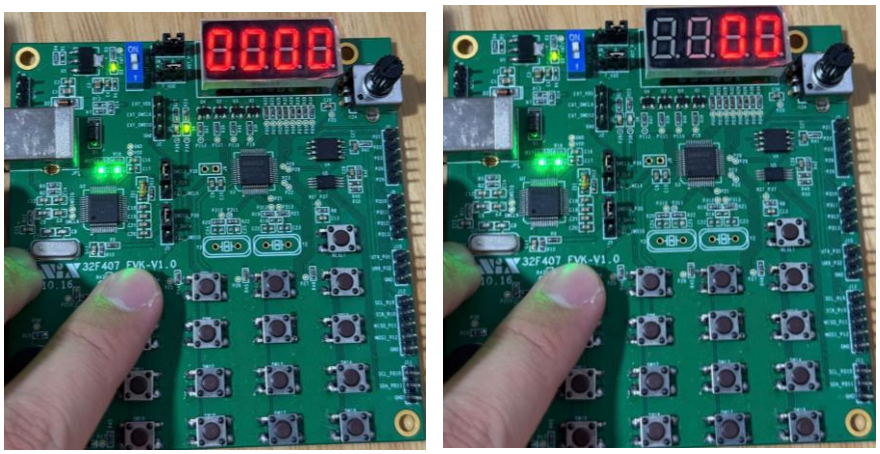




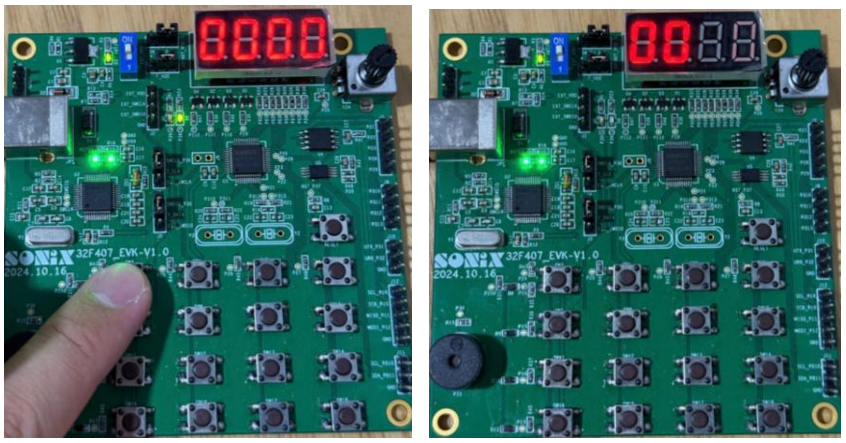


Requirement 2: SETUP BUTTON (SW3):

- While in normal mode, press SW3 to enter time-setting mode: the HH 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).



- While in hour-adjustment mode, press SW3 to enter minute-adjustment mode: the MM 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

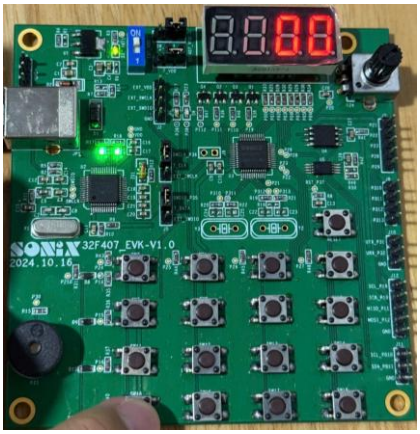


- While in minute-adjustment mode, press SW3 to enter normal mode.

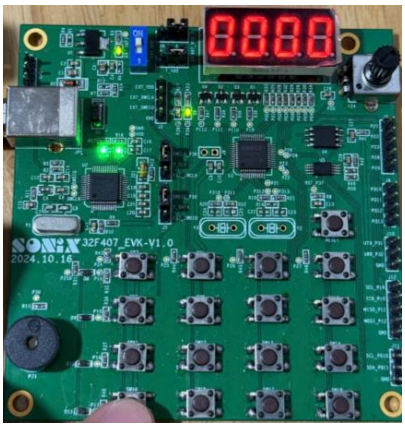


Requirement 3: TIMER BUTTON (SW16):

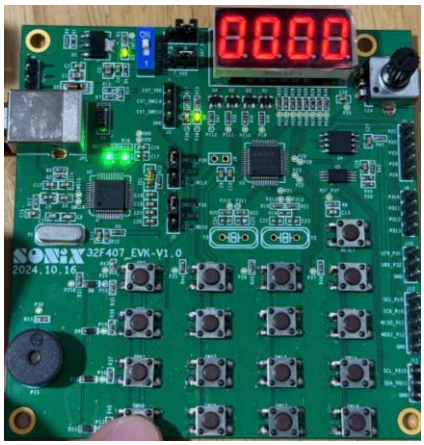
- While in normal mode, press SW16 to enter the scheduled time adjustment mode: the HH 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

- While in the scheduled time adjustment mode, press SW16 to enter the scheduled minute adjustment mode: the MM 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

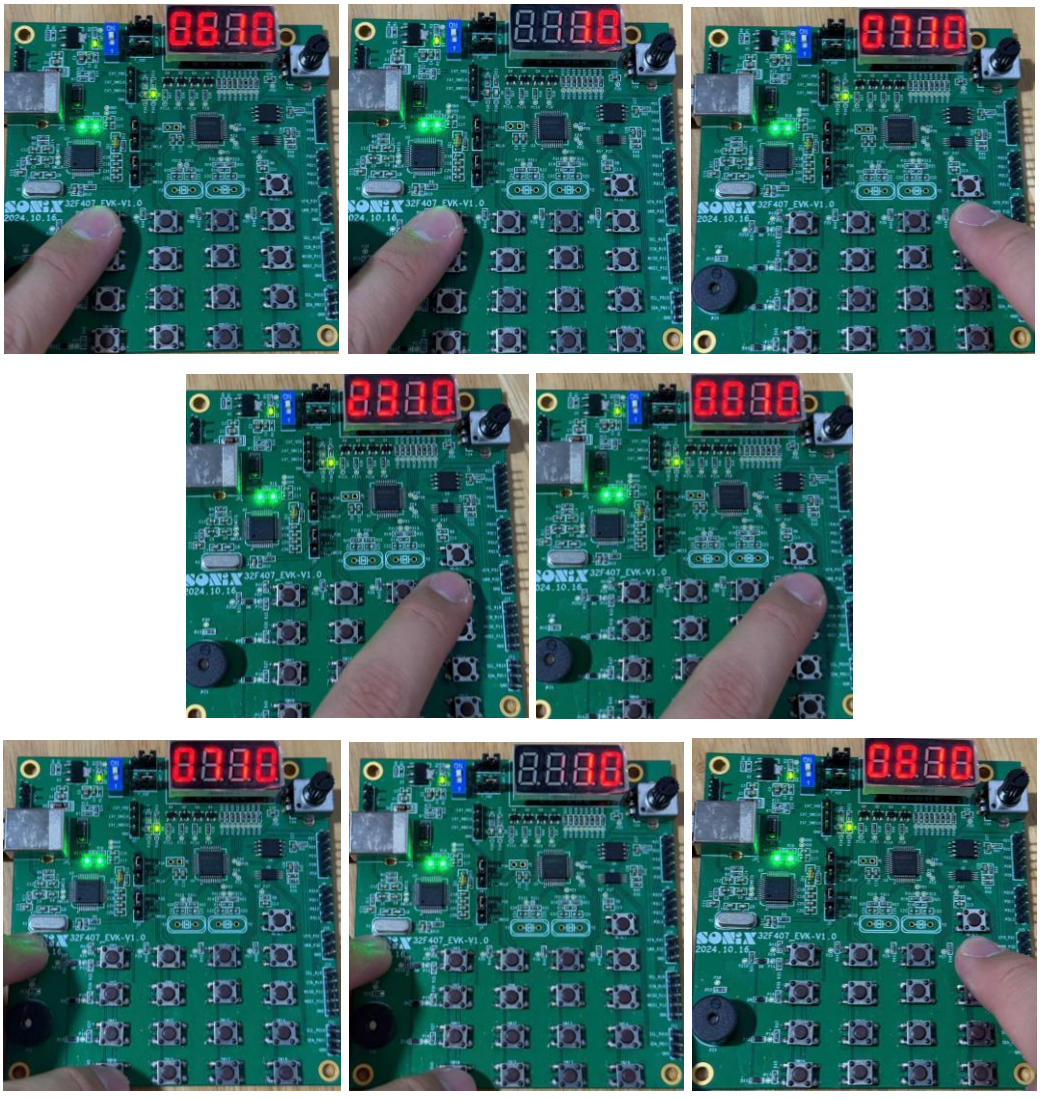


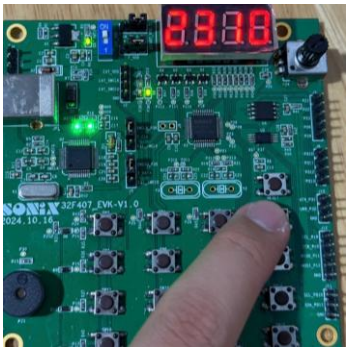

- While in the appointment minute adjustment mode, press SW16 to save the appointment time to EEPROM and exit to normal mode.



Requirement 4: PLUS BUTTON (SW6):

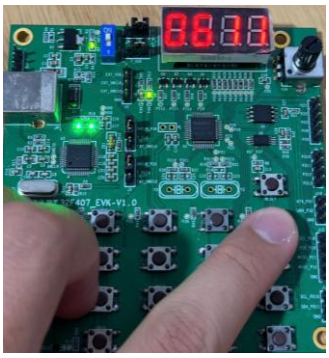
- While in time-setting or alarm-setting mode, press SW6 to increment the hour by 1; the hour resets to 0 upon reaching 24.





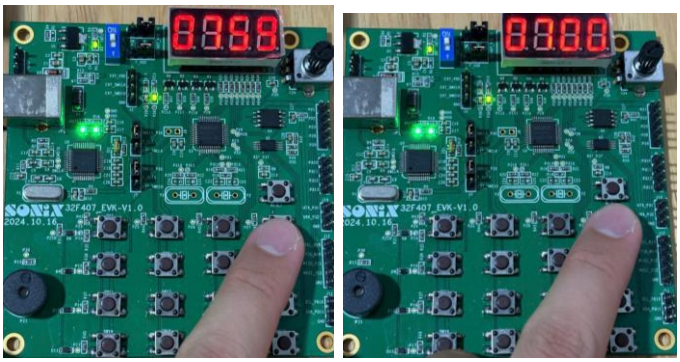

- While in minute adjustment mode or alarm minute adjustment mode, press SW6 to increment the minutes by 1; when the minutes reach 60, they will reset to 0.





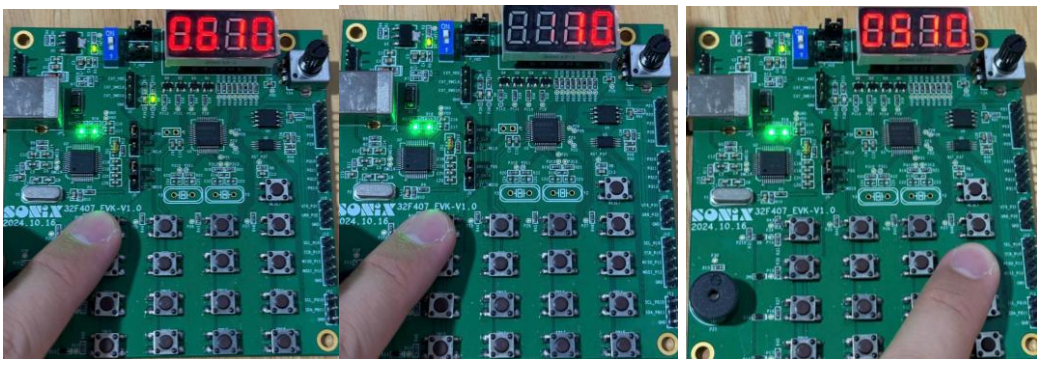


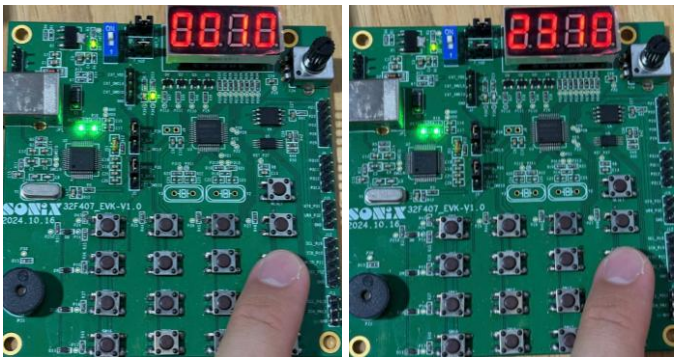


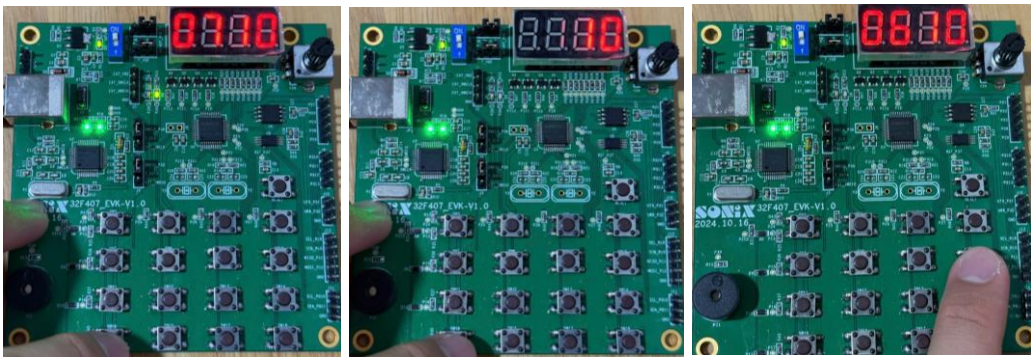



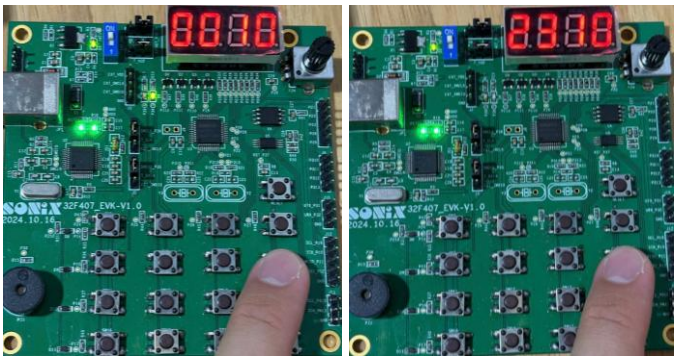
Requirement 5: BUTTON – (SW10):

- In time-setting or alarm-time-setting mode, press SW10 to decrease the hour by 1; if the hour is 0, decreasing it by 1 sets it to 23.

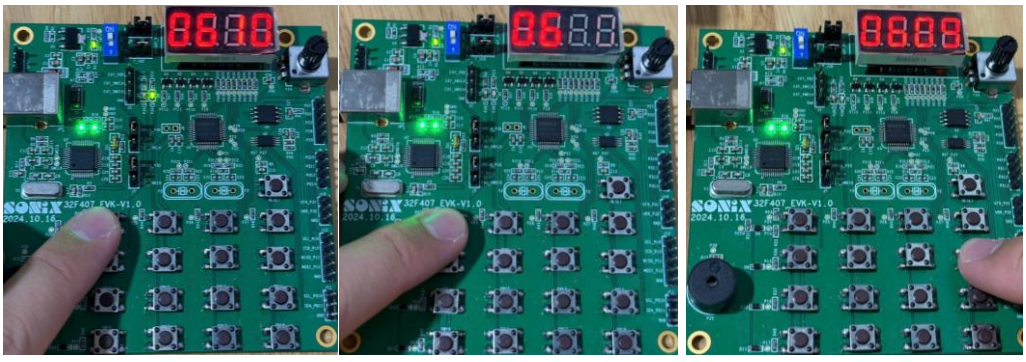


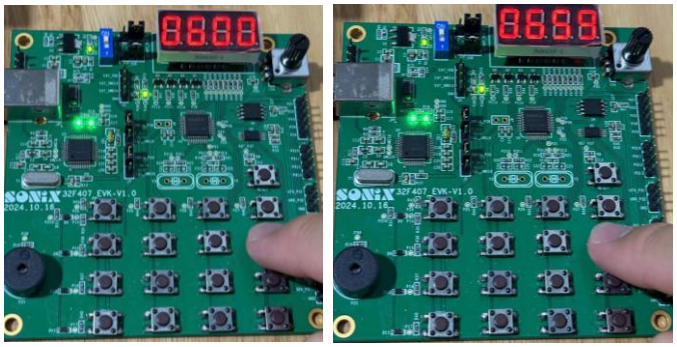


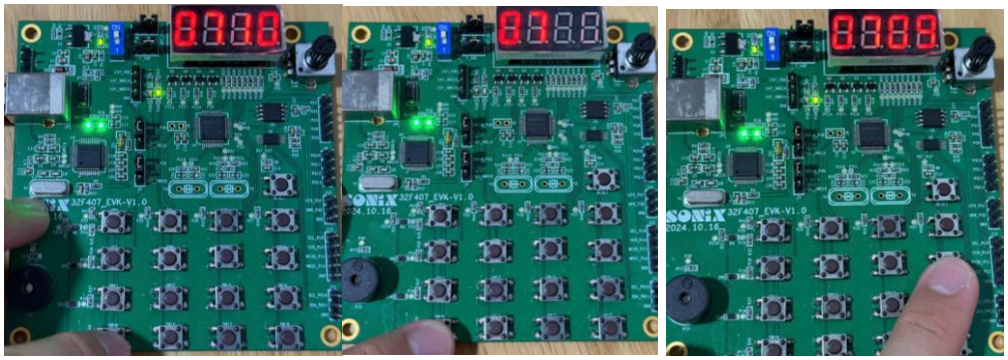


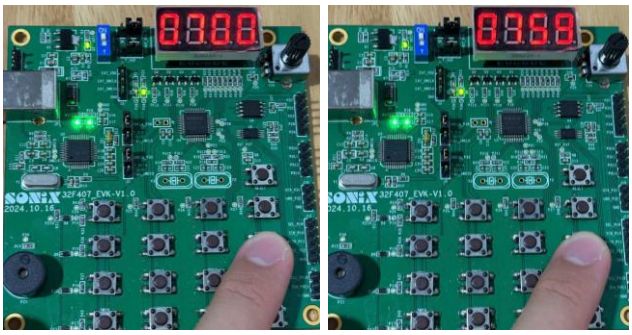


- In minute-setting or alarm-minute-setting mode, press SW10 to decrease the minute by 1; if the minute is 0, decreasing it by 1 sets it to 59.









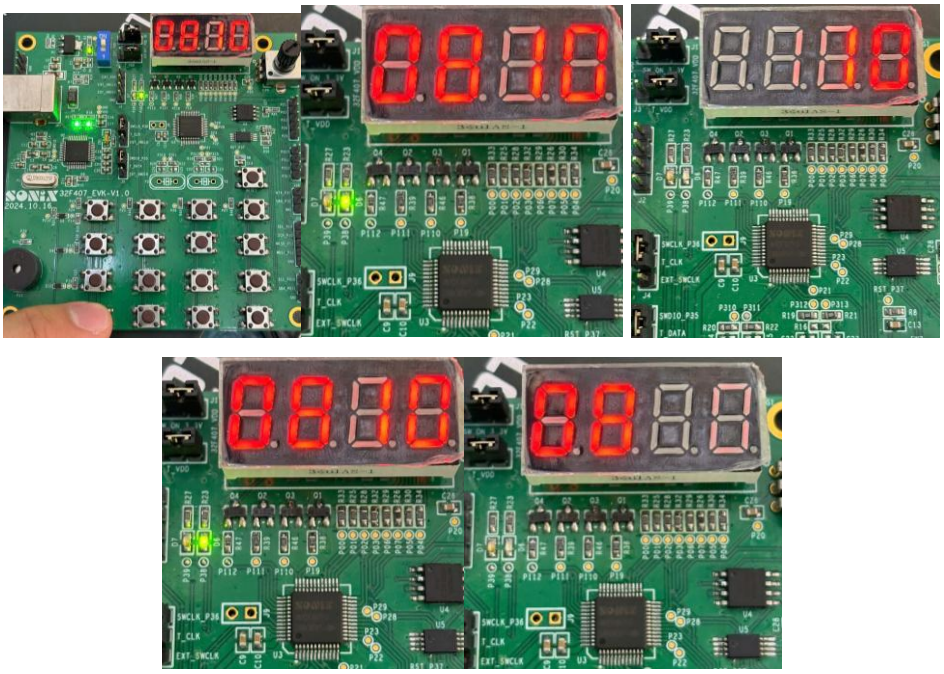
Requirement 6: BUZZER:

- Pressing buttons SW3, SW6, SW10, or SW16 triggers a 0.3s beep.
- When the clock reaches the scheduled alarm time, the buzzer emits a continuous "beep-beep" pattern for 5 seconds (0.5s ON – 0.5s OFF cycle).
- Upon timing out and exiting the hour/minute adjustment mode or the alarm hour/minute adjustment mode to return to normal mode, the buzzer emits a 0.3s beep.

(See the results in the demo video)

Requirement 7: LED (D6 ON SN32F407_EVK BOARD):

- In alarm hour or alarm minute adjustment modes, the LED flashes with a 1s cycle (0.5s ON – 0.5s OFF). Otherwise, the LED remains OFF.



Requirement 8: EEPROM:

- Stores the scheduled alarm hour and minute; default alarm seconds = 0.

(See the results in the demo video)

Requirement 9: TIMEOUT ON NO BUTTON ACTIVITY:

- In hour/minute adjustment mode or alarm hour/minute adjustment mode, if no button is pressed within 30 seconds, the system exits to normal mode.

(See the results in the demo video)

6 References

[1] SONiX Technology Co., Ltd., *SONiX SN32F407 EVK User Guide*, Version 1.0, 2026.

[2] SONiX Technology Co., Ltd., *SN32F400 Series User's Manual – SN32F407, SN32F405, SN32F403*, Version 2.2, 2025.

[3] John Tsai (MCU FAE), *Introduction of SONiX SN32F407 EVK Kit*, SONiX Technology Co., Ltd., 30/07/2026.